

Per Lane Id detector Functional Design Specifications

UCle 3.0 Physical layer

Version 1

Block Owner

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2 Revision History

Version	Date	Author(s)	Revision Notes	Owner Approval

3 Overview

This block is mainly used to detect if the lane id pattern is detected successfully or not

4 Operation and Description

4.1 Digital Interface

4.1.1 Parameters Names

Parameter Name	Default	Description
pDATA_WIDTH	64	It defines the width of the data input port
pLANE_ID_PATTERN	16	It says the pattern specified per lane

4.1.2 Ports Names

Port Name	Port Width	Port Type	Description
i_data_in	pDATA_WIDTH	input	It is the data input needed to be compared
i_clk	1	Input	The clock used in the design
i_enable	1	Input	Enable for the detection
i_reset_n	1	input	Reset of the system
o_laneid_success	1	output	Signal is asserted if the pattern is successful

4.2 Functional Description

- This block is mainly used to detect if 16 consecutive pattern of lane id pattern is received or not and sends a signal to the RX controller indicating if the lane is good or not
- Pattern register is a read-only register that stores the lane id pattern concatenated 4 times ({16'b lane_id, 16'b laneid}).

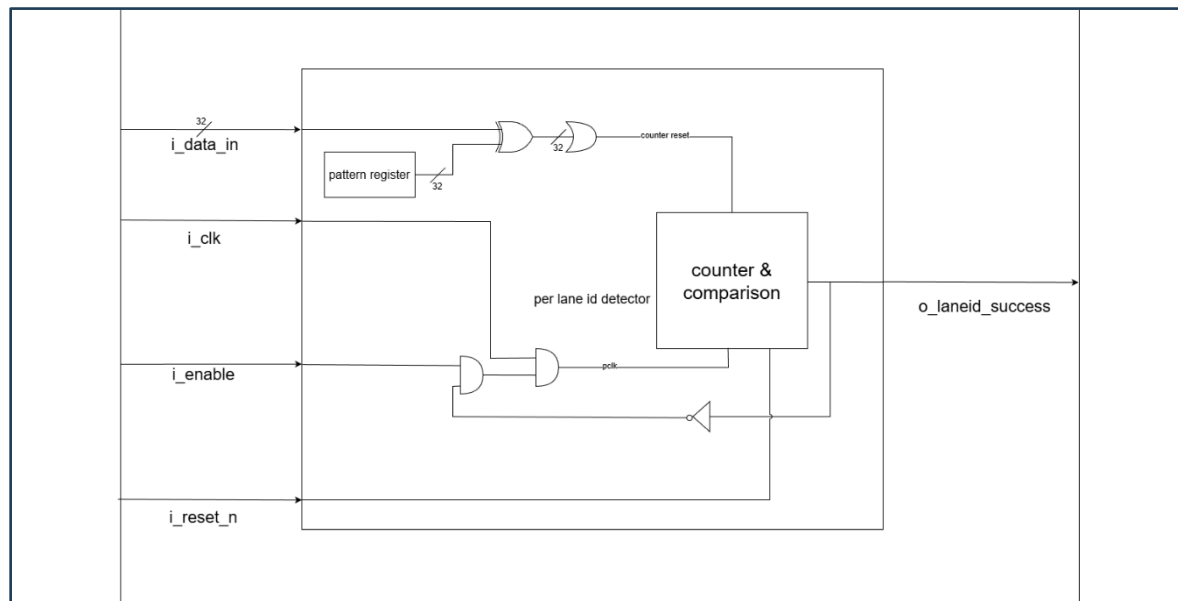
Per Lane ID Pattern																
Pattern	0	1	0	1	Lane ID ^a								0	1	0	1
Bit	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

a. Note that bit 0 of Lane ID maps to bit 4 in the Per Lane ID pattern, bit 1 to bit 5 and so on until bit 7 to bit 11.

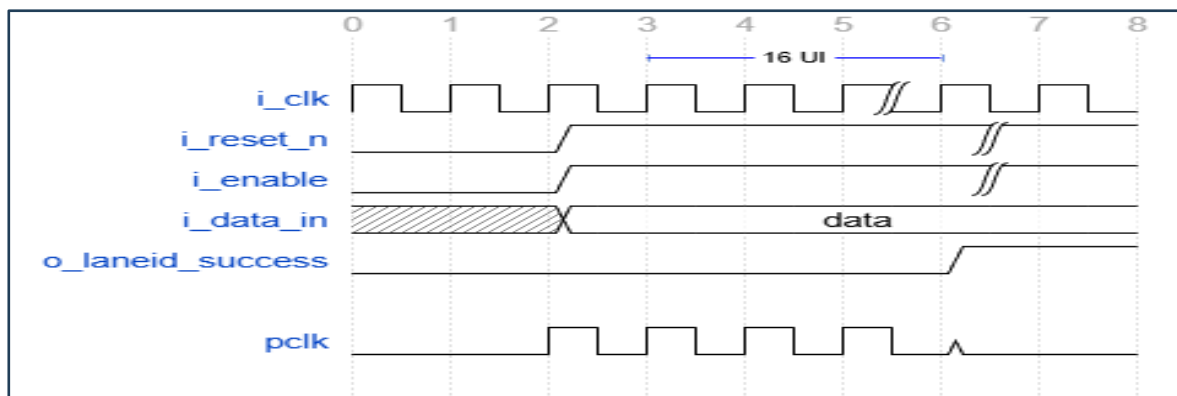
Logical Lane Number	Lane ID	Logical Lane Number	Lane ID
0	00000000b	8	00001000b
1	00000001b	9	00001001b
2	00000010b	10	00001010b
3	00000011b	11	00001011b
4	00000100b	12	00001100b
5	00000101b	13	00001101b
6	00000110b	14	00001110b
7	00000111b	15	00001111b

- The local pattern is compared with the incoming pattern and if an error is detected it will reset the counter
- The counter counters every cycle but resets if i_reset_n is asserted or an error is detected
- If the counter increments 16 cycles consecutively it will assert o_laneid_success and stop the counting and detection

4.3 Block Diagram



4.4 Timing Diagram



4.5 Verification Requirements

- 4.5.1 Test if the incoming data is the same as the pattern saved in the register
- 4.5.2 Test if successful 16 consecutive pattern is detected successfully the `o_lane_success` is asserted and the counter stop counting
- 4.5.3 If a pattern is detected wrong the counter is reset and back to zero or reset was asserted
- 4.5.4 The counter only counts if the enable and no lane success